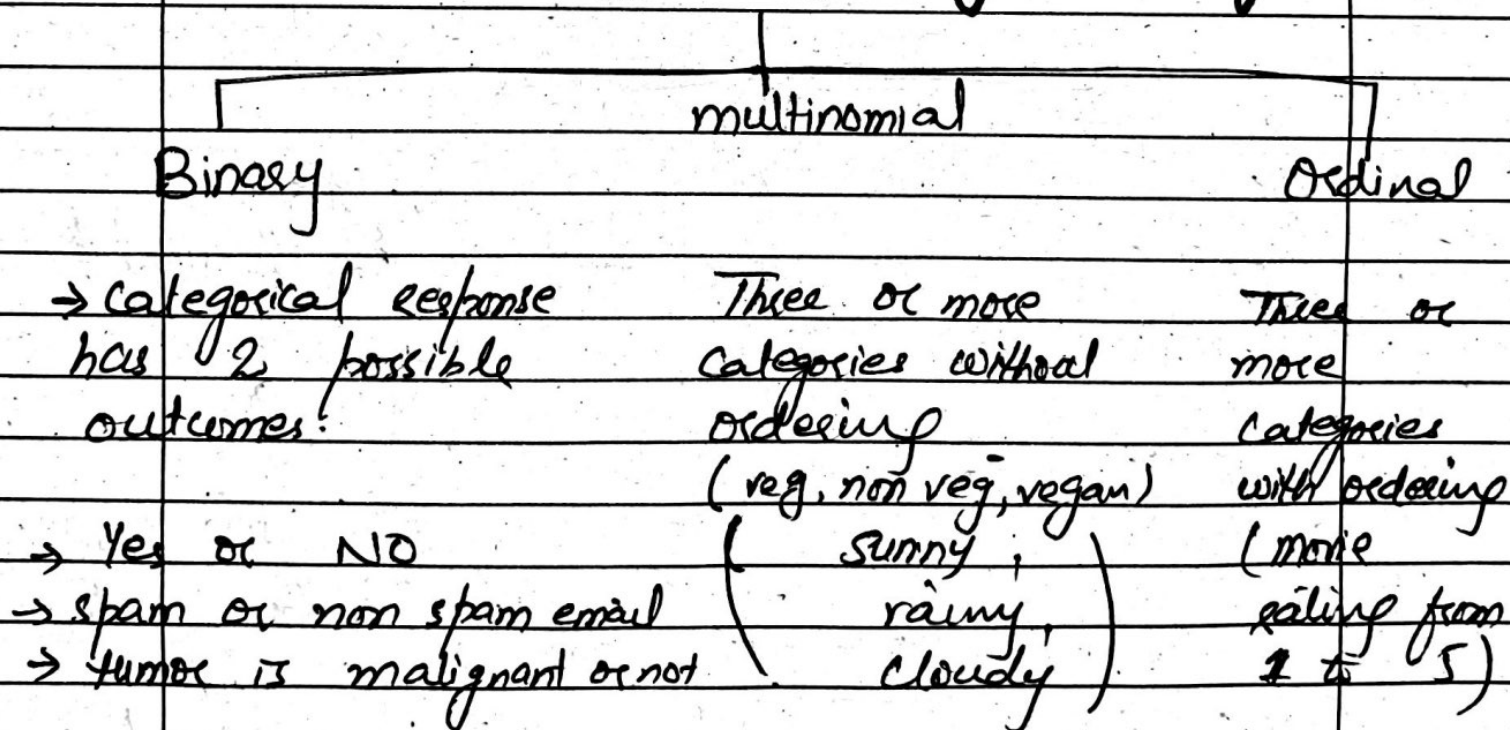


Logistic Regression

- also called logistic/logit model.
- analyze the relationship b/w multiple independent and a categorical dependent variable (Y)
- it predict the probability of a categorical dependent variable.
- it deal with discrete value.

Types of logistic Regression

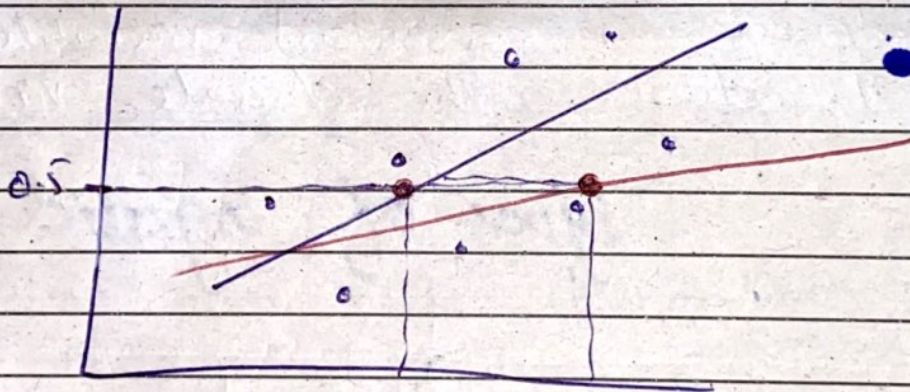


Why we used these problem through linear regression.

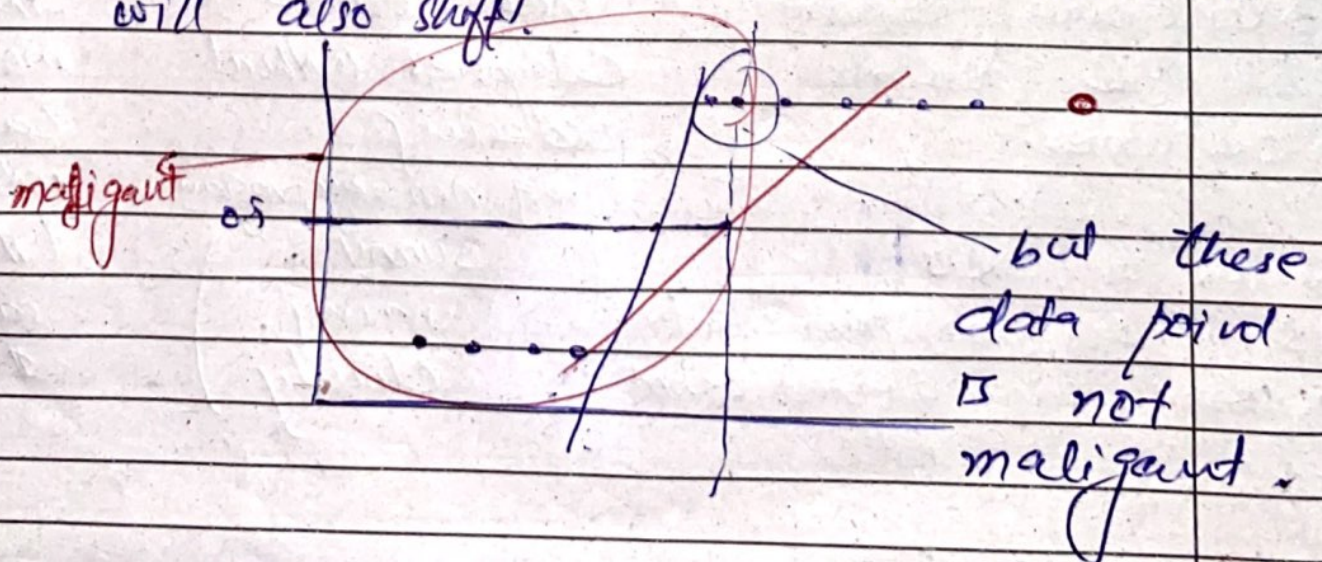
① In linear regression output will come in continuous, its range lies $-\infty$ to ∞ .

but we want values range (0 to 1), that we can't get from linear regression.

② Suppose we had few data point, we plot line



if new data point add, the line will change. ~~and~~ and threshold will also shift.



Sigmoid function

→ is used to calculate the probability.

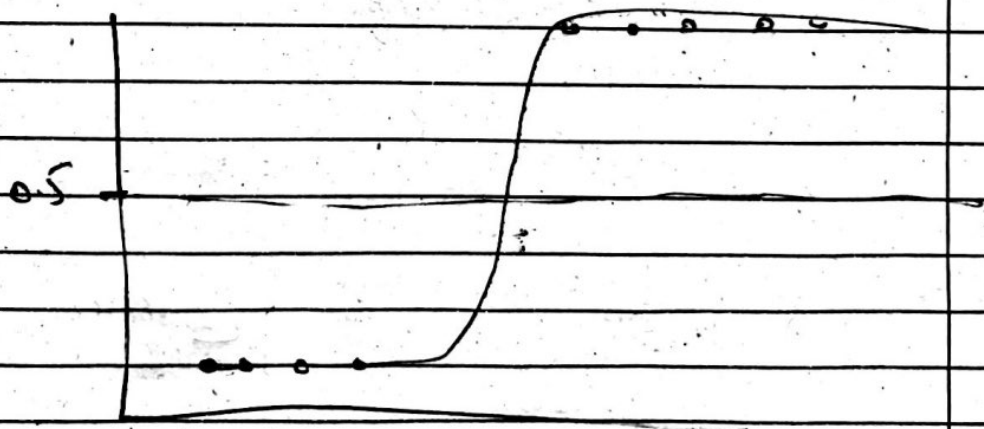
$$f(x) = \frac{1}{1 + e^{-x}}$$

e = the natural logarithm base
(also known as Euler's number)

" e " is mathematical constant.
that is the base of the natural logarithm.

$$e = 2.71828$$

→ also called S shaped curve.



→ logistic regression give output either "1" or "0"

→ input can be anything, it lies from $-\infty$ to ∞ .

ODDS

→ are the ratio of the probability that the event will occur to the probability that it will not occur.

$$\text{ODDS}[\text{event}] = \frac{p}{1-p} \quad \begin{array}{l} \text{so } p \text{ event occurs} \\ \text{in } 1-p \text{ event can't} \\ \text{occur.} \end{array}$$

it ranges from "0" to ∞

Taking natural log

$$= \log \left(\frac{p}{1-p} \right)$$

it ranges from $-\infty$ to ∞
same as linear regression.

$$\log \left(\frac{p}{1-p} \right) = \alpha + \beta x$$

$$\frac{p}{1-p} = e^{(\alpha + \beta x)}$$

Taking inverse

$$\frac{1-p}{p} = \frac{1}{e^{(\alpha + \beta x)}}$$

$$\frac{1}{p} - \frac{p}{p} = \frac{1}{e^{(\alpha + \beta x)}}$$

$$\frac{1}{p} - 1 = \frac{1}{e^{(\alpha + \beta x)}}$$

$$\frac{1}{p} = \frac{1}{e^{(\alpha + \beta x)}} + 1$$

$$\frac{1}{p} = \frac{1 + e^{(\alpha + \beta x)}}{e^{\alpha + \beta x}}$$

$$p = \frac{e^{(\alpha + \beta x)}}{1 + e^{(\alpha + \beta x)}}$$

$$= \frac{1}{e^{-(\alpha + \beta x)} (1 + e^{(\alpha + \beta x)})}$$

$$= \frac{1}{e^{-(\alpha + \beta x)} + e^{-\cancel{(\alpha + \beta x)} + (\alpha + \beta x)}}$$

$$= \frac{1}{e^{-(\alpha + \beta x)} + e^0}$$

$$= \frac{1}{e^{-(\alpha + \beta x)} + 1}$$

$$p = \frac{1}{1 + e^{-(\alpha + \beta x)}}$$

Cost function

$$\text{Cost} (h_{\theta}(x), y_{\text{actual}})$$

$$= \begin{cases} -\log(h_{\theta}(x)) & y=1 \\ -\log(1-h_{\theta}(x)) & y=0 \end{cases}$$

$$= -y \log(h_{\theta}(x)) - (1-y) \log(1-h_{\theta}(x))$$

$$\therefore h_{\theta}(x) = \hat{y}$$

$$= -y \log \hat{y} - (1-y) \log(1-\hat{y})$$

$$= -(y \log \hat{y} + (1-y) \log(1-\hat{y}))$$

$$= -L(\hat{y}, y)$$

$$-L(\hat{y}, y) \quad \therefore \hat{y} = a = \frac{1}{1+e^{-z}}$$

$$\therefore z = mx + c$$

$$\frac{\partial L}{\partial m} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial m}$$

$$\frac{\partial L}{\partial a} = \frac{\partial}{\partial a} (-L(\hat{y}, y))$$

$$= \frac{\partial}{\partial a} (-y \log \hat{y} - (1-y) \log(1-\hat{y}))$$

$$\therefore y = a$$

$$= \frac{\partial}{\partial a} (-y \log a + (1-y) \log(1-a))$$

$$= -y \left(\frac{1}{a} \right) + (1-y) \left(\frac{1}{1-a} \right) (-1)$$

$$\boxed{\frac{\partial L}{\partial a} = -\frac{y}{a} + \frac{1-y}{1-a}}$$

$$\frac{\partial a}{\partial z} = \frac{\partial}{\partial z} \left(\frac{1}{1+e^{-z}} \right)$$

$$= \frac{\partial}{\partial z} (1+e^{-z})^{-1}$$

$$= -1 (1+e^{-z})^{-2} (e^{-z} (-1))$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$$

$$\boxed{\frac{\partial a}{\partial z} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}}$$

$$\therefore a = \frac{1}{1+e^{-z}}$$

$$\therefore 1-a = 1 - \frac{1}{1+e^{-z}}$$

$$= \frac{1+e^{-z} - 1}{1+e^{-z}} = \frac{e^{-z}}{1+e^{-z}}$$

$$\therefore 1-a = \frac{e^{-z}}{1+e^{-z}}$$

$$\frac{\partial z}{\partial m} = \frac{\partial}{\partial m} (mx + c)$$

$$\boxed{\frac{\partial z}{\partial m} = x}$$

$$\frac{\partial L}{\partial m} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial m}$$

$$= \left(-\frac{y}{a} + \frac{1-y}{1-a} \right) (a(1-a)) (x)$$

$$= \left(\frac{-y(1-a) + a(1-y)}{a(1-a)} \right) (\cancel{a(1-a)}) (x)$$

$$= (-y + \cancel{ay} + a - \cancel{ay}) (x)$$

$$\boxed{\frac{\partial L}{\partial m} = (a - y) (x)}$$

$$\frac{\partial L}{\partial c} = \frac{\partial L}{\partial a} \cdot \frac{\partial a}{\partial z} \cdot \frac{\partial z}{\partial c}$$

$$= \left(-\frac{y}{a} + \frac{1-y}{1-a} \right) (a(1-a)) (1)$$

$$\boxed{\frac{\partial L}{\partial c} = (a - y)}$$

$$m_{\text{new}} = m - \eta \cdot \frac{\partial L}{\partial m}$$

$$c_{\text{new}} = c - \eta \cdot \frac{\partial L}{\partial c}$$

Question

X	y	Cost	$\frac{1}{1+e^{-(mx+c)}}$	$a-y$	$(a-y)(z)$
0.50	0	0.301	0.5	0.5	0.25
0.75	1	0.301	-0.5	-0.5	-0.375
1.00	0	0.301	0.5	0.5	0.5
1.25	0	0.301	0.5	0.5	0.625
1.50	1	0.301	-0.5	-0.5	-0.75
1.75	1	0.301	-0.5	-0.5	-0.875
1.505	0		0		+0.625

$$m=0.03125$$

$$c=0$$

$$\hat{y} = \frac{1}{1+e^{-(mx+c)}}$$

$$0.5039 \approx 1$$

$$0.5058 \approx 1$$

$$0.5078 \approx 1$$

$$0.5097 \approx 1$$

$$0.5117 \approx 1$$

$$0.5136 \approx 1$$

Step I initialization

$$m=0 \quad c=0$$

$$\text{threshold} = 0.5$$

$$\text{learning rate} = 0.05$$

Step II Cost

$$= \frac{1}{n} \sum_{i=1}^n [-y \log \hat{y} - (1-y) \log (1-\hat{y})]$$

$$\hat{y} = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{5} (1.505)$$

$$\approx 0.301$$

$$z = mx + c$$

Step III direction

$$\frac{\partial L}{\partial m} = \sum_{i=1}^n (a - y_i)(x_i)$$

$$= (-0.625)$$

$$= 0.000$$

$$\frac{\partial L}{\partial c} = \sum_{i=1}^n (a - y_i)$$

$$= 0$$

Step IV New value

$$m_{\text{new}} = m - lr * \frac{\partial L}{\partial m}$$

$$= 0 - 0.05 * -0.625$$

$$= +0.00625 = 0.03125$$

$$c_{\text{new}} = c - lr * \frac{\partial L}{\partial c}$$

$$= 0 - 0.05 * 0$$

$$= 0$$

Step v predicted

$$= \frac{1}{1 + e^{-(mx+c)}}$$

$$\therefore m = 0.03125$$

$$\therefore c = 0$$